

ANUSHKA SUNIL

Email: anushkasunil71@gmail.com | anushka.sunil@nitdelhi.ac.in

ORCID ID: <https://orcid.org/0009-0002-6293-0556>

LinkedIn ID: <https://www.linkedin.com/in/anushka-sunil-579115317/>

Professional Summary

As a climate researcher, I study the impacts of climate change and natural variability on Arctic sea ice to understand the mechanisms driving polar climate variability. My research integrates satellite observations and reanalysis datasets (ERA5, NSIDC etc.) to investigate atmosphere-ocean-ice interactions and their influence on sea ice dynamics. By employing advanced analytical methods, I aim to deepen understanding of Arctic climate sensitivity and its connections to global circulation systems, contributing to more reliable projections of polar climate responses in a warming world.

Academic Background

Degree/Certificate	University/Board	Percentage/CGPA	Year
Ph.D. (Mathematics)	National Institute of Technology Delhi	9.44 CGPA	2023-Present
M.Sc. (Mathematics)	Dayalbagh Educational Institute, Agra	8.97 CGPA	2020-2022
D.El.Ed.	P.N.P. Allahabad, Prayagraj	90.15%	2018-2020
B.Sc. (PCM)	M.J.P. Rohilkhand University, Bareilly	73.18%	2015-2018
Senior Secondary	U.P. Board	76.20%	2015
Secondary	U.P. Board	81.83%	2013

Publications

- Sunil, A.**, Kumar, P., Singh, A., Agarwal, A., Kumar, A., Kim, S. Y., & Rajni. (2025). Assessing the Impact of Climate Modes on Extreme Arctic Sea Ice Using Reanalysis Data. *International Journal of Climatology*, 45(15), e70121.
- Sunil, A.**, Kumar, P., Singh, A., Agarwal, A., Kumar, A., Kim, S. Y., & Rajni. Indian Ocean Sea Surface Temperature Modulates the Arctic Oscillation: A Multi-Model Assessment Using CMIP6. *International Journal of Climatology*, e70259.

Conference Proceedings

- Sunil, A.**, Kumar, P., & Kumar, A. (2025). Role of Major Climate Modes on Arctic Sea Ice Dynamics. *National Conference on Polar Sciences 2025*, NCPOR Goa, September 2025. (Poster)

Achievements and Awards

- Qualified GATE 2023 (Mathematics).
- Secured 1st rank in M.Sc. (Mathematics) and 2nd overall among candidates of M.Sc. Mathematics / M.Sc. Mathematics with specialization in Computer Science, Dayalbagh Educational Institute, Agra.
- Secured 2nd rank in B.Sc. (Mathematics), N.M.S.N. Dass College (Affiliated to Rohilkhand University).

Technical Skills and Certifications

- **Software:** Origin, LaTeX, Simulink, GrADS, CDO, Microsoft Office (Word, Excel, PowerPoint)
- **Programming Languages:** MATLAB, Python, Fortran, C/C++, Java
- **Soft Skills:** Teamwork, Leadership, Event Management, Written and Oral Communication
- **Certification:** Course on Computer Concepts (CCC)

Internships & Workshops

- **Natural Hazard Symposium for the Indian Himalayas 2024, IIT Roorkee** (October 8-10, 2024) - Participated in discussions and technical sessions focused on natural hazards, climate variability, and risk assessment in the Himalayan region.
- **Workshop on “MATLAB for Teaching & Research”, NIT Delhi** (August 21-23, 2024) - Gained hands-on experience in MATLAB for academic teaching and research applications, including best practices and practical implementation.
- **5-Day Workshop on Topics in Generative AI, NIT Delhi** (July 20-24, 2024) - Participated in an intensive workshop covering fundamentals and applications of Generative AI.
- **Software Development Intern, EXPOSYS DATA LABS**, (Jun-Jul 2023) - Developed an E-learning website platform, contributing to frontend design, functionality implementation, and content structuring.
- **FDP on Differential Equations**, Division of Mathematics, School of Advanced Sciences, VIT Chennai (June 12-16, 2023) - Completed a five-day online FDP emphasizing analytical and applied aspects of differential equations.

Extra-Curricular Activities

- **Anchored the 4th IEEE International Conference TEMSMET 2025**, organized by NIT Delhi, where I introduced eminent speakers and moderated sessions, strengthening professional communication, stage presence, and audience engagement skills.
- **Secured 3rd Position in a District-Level Speech Competition**, demonstrating strong public speaking, articulation, and confidence.
- **Participated in National Service Scheme (NSS) Camp**, actively contributing to community development and social outreach activities.
- **Participated in Scout & Guide Camp**, fostering teamwork, leadership, discipline, and collaborative skills.

References

Dr. Prashant Kumar

Associate Professor, Department of Applied Sciences (Mathematics)
National Institute of Technology Delhi, India

Email: prashantkumar@nitdelhi.ac.in

Relationship: Ph.D. Supervisor

Dr. Anurag Singh

Associate Professor, Department of Computer Science & Engineering
National Institute of Technology Delhi, India

Email: anuragsg@nitdelhi.ac.in

Relationship: Ph.D. Co-Supervisor